

SHETH LUJ COLLEGE

CYBER SECURITY PRACTICAL - 4

Aim:-Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CODE:-

```
CS_Practical4.py - D:/daniyal/CS_Practical4.py (3.14.6)
File Edit Format Run Options Window Help

import hashlib
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes

# Generate RSA private key
private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)

# Get public key
public_key = private_key.public_key()

# Take message from user
message = input("Enter message: ")

# Convert message into bytes
message_bytes = message.encode()

# Create digital signature using private key
signature = private_key.sign(
    message_bytes,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

print("\nOriginal Message:", message)
print("Digital Signature:", signature.hex())

# Verify the digital signature
try:
    public_key.verify(
        signature,
        message_bytes,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("\nSignature Verification: SUCCESS")
    print("Message is authentic and has not been modified.")
except Exception:
    print("\nSignature Verification: FAILED")
    print("Message has been modified or signature is invalid.")

# Test message integrity
modified_message = input("\nEnter modified message to test: ")

try:
    public_key.verify(
        signature,
        modified_message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("Modified Message Verification: SUCCESS")
except Exception:
    print("Modified Message Verification: FAILED")
    print("The message has been changed, so authenticity/integrity is not valid.")
```

```
CS_Practical4.py - D:/daniyal/CS_Practical4.py (3.14.6)
File Edit Format Run Options Window Help

    ), salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

print("\nOriginal Message:", message)
print("Digital Signature:", signature.hex())

# Verify the digital signature
try:
    public_key.verify(
        signature,
        message_bytes,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("\nSignature Verification: SUCCESS")
    print("Message is authentic and has not been modified.")
except Exception:
    print("\nSignature Verification: FAILED")
    print("Message has been modified or signature is invalid.")

# Test message integrity
modified_message = input("\nEnter modified message to test: ")

try:
    public_key.verify(
        signature,
        modified_message.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("Modified Message Verification: SUCCESS")
except Exception:
    print("Modified Message Verification: FAILED")
    print("The message has been changed, so authenticity/integrity is not valid.")
```

SHETH LUJ COLLEGE

CYBER SECURITY PRACTICAL - 4

OUTPUT:-

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/daniyal/CS_Practical4.py =====
Enter message: My name is Ankita

Original Message: My name is Ankita
Digital Signature: 1c8404eed9f4393178465a0a955a1910d4b4ef001134469aa61ddb95ecb3cd55fd182790959ca67fc87f769a869ebb9bdc9a1a9e9cfa2be1201af282bd2908802061e5c3b7f7642d24f36
89f16ef5e0b83b132841008a2f0bb60fec0e81d0d1a9590f1fe8b1941fd2397bd75d185a92bab5b199056570a2cd4cfd6a4024a4efcd1698ae0542714d02355b7cf45ba8bb2689f189ac75e662d2ce9b4b159a0b5a
71a703a09927d59725a9a66075b39baa472d297bfcae5c62c9a7e5a030095f605a04eaf487f839bfc9f63alc5cf319be9dcccddade1e5bc6d62a9f2bd7c68e52a299d935adf1b6b79880823d30d731f93eca7622f95
a6fbfe19937262c7e9525

Signature Verification: SUCCESS
Message is authentic and has not been modified.

Enter modified message to test: My name is not ankita
Modified Message Verification: FAILED
The message has been changed, so authenticity/integrity is not valid.

>>>
===== RESTART: D:/daniyal/CS_Practical4.py =====
Enter message: My name is Ankita

Original Message: My name is Ankita
Digital Signature: a71a3464059f35af7efddca29473574cbdb97502f7035e1f4481247e1cb249db6a0c891118b0a1285e224f875ab5395423ae807bf55d039a8ae45cb211e427f92c6eba41bf4beb8a381ecc
9826108aef1b13855ed13b931b1b9650444b4b53c114d95180f98c6092e77d6ca74067021581859e4f68d849d49d4214ea1468a27eb302ba602833cd252df777314bd059e53a27fd4b07bfb93d6f6b73268da43668
0aba9190d0fe3b1c0e9b91a8fcad5a50fb870077743c55c591736b53399daa54945c7a480a63a98a2f2d6caa187fa08012d846e783232e306f0cda6f388ffa8295134b3c91199f9f68630ceff6e38eb9bb6a9ddcee
a903f6046af51fcf2ff10

Signature Verification: SUCCESS
Message is authentic and has not been modified.

Enter modified message to test: My name is Ankita
Modified Message Verification: SUCCESS

>>>
```